



losequenceIMSLDBridge

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About

This document details the work done for the losequenceIMSLDBridge project which is a proof-of-concept project for exporting levelA IMS LD from the activity sequences built using the phosphorix IOPortal technology.

A proof-of-concept levelA IMS LD export was developed for activity sequences built using IOPortal technology. The deliverables for the project are as follows:

1. A proof-of-concept prototype for automatically exporting levelA IMS LD from activity sequences built using IOPortal technology.
2. An Open Source Java API for managing levelA IMS LD. The code for these API have been released under Apache 2.0 license by Phosphorix Ltd.
3. Javadoc and an example java file demonstrating using these java api for levelA IMS LD.
4. Report in word format containing the technical overview and the screen shots of the working prototype. As part of the report the data transformation mapping between ioSequence and IMS LD have also been created.

The specific output files from the project are as follows:

- 1.losequenceIMSLDBridge Main Report.pdf (.doc and .swx versions also): This report contains the technical overview of the working prototype.
- 2.losequenceIMSLDBridge Prototype Screen shots.pdf: The file contains the screen shots of the working prototype.
- 3.losequence_IMSLD_Mapping.xls: This file contains the data transformation mapping between ioSequence and IMS LD.
- 4.losequence XSD Schema.xls: This file contains description for the various fields in the ioSequence specification.
- 5.IMSLDV1.jar: This is an Open Source Java API for managing levelA IMS LD developed by Phosphorix Ltd.
- 6.javadoc for imslld level a api.zip: This file contains the javadoc for the Open Source Java API for managing levelA IMS LD.
- 7.Source for Java API for LevelA IMS LD.zip: This file contains the source code for the Open Source Java API for managing levelA IMS LD. This code has been released under Apache 2.0 license by Phosphorix Ltd.
- 8.TestIMSLDAutoGenerateAPI.java: This is an example file containing an example to show the usage of the Open Source Java API for managing levelA IMS LD.



Background

LD4P, reload, openPDD, icebox and ioPortal project [x many] will benefit from being able to export or import personal development activities and learning activities.

Phosphorix use ioSequence xml to map out and store browser based activities.

The ioSequence player is used in ioPortal for things like CV building, activity, web site building activity and recently Q & A surveys etc.

Mark Baxendale and Selwyn noted similarities and possibilities for a "personal development design" approach to creating activities that is not unlike the learning design initiatives and project approaches used in Coppercore and Reload.

ioMorph allows data transformation between xml types and simple text types.

The challenge for this project is to do a data transformation between ioSequence XML and IMS Learning Design.

Furthermore, the challenge is also to demonstrate an activity designed in Liverpool Hopes "icebox" being loaded into the Reload designer and perhaps tested in an appropriate player.

Similarly it might also be possible to load an IMS LD file into ioPortal and play it in the Icebox portal, however this may involved extending ioPortal and ioSequence xml and the output instead would be a specification of work required.

Scope

We should be able to create a demonstrator which can show the benefits of an interoperable approach to the wider LD and PDP communities and hopefully illustrate the synergy between learning design and PDP activities.

Deliverables

1. A proof-of-concept working prototype for automatically exporting levelA IMS LD from activity sequences built using IOPortal technology.
2. An Open Source Java API for managing levelA IMS LD. The code for these API have been released under Apache 2.0 license.
3. Javadoc and an example java file demonstrating using these java api for levelA IMS LD.
4. Report in word format containing the technical overview and the screen shots of the working prototype. As part of the report the data transformation mapping between ioSequence and IMS LD have also been created.



losequence XSD

The losequence specification has been developed by phosphorix and is being used in the ioportal for creating activities.

The schema of the losequence specification has been described in the file losequence XSD Schema.xls.

IMS LD Standard

The IMS LD [1] specification contains the data models to represent the assessments and results.

IMS Ld are basically a collection of components and method. Components include a collection of roles, activity, environments (tools, services like email, communication) Methods assign activity to roles.

UoL (Unit of Learning) is basically a bundle of Ld, resources and physical files. Formally UoL is a IMS Content Package.

The latest version of this specification is the version 1.0 which can be accessed at

<http://www.imsglobal.org/learningdesign/>

The following document provides an overview of this specification and can be a good starting point to understand the learning design:

- IMS Learning Design Best Practice and Implementation Guide,
http://www.imsglobal.org/learningdesign/ldv1p0/imslld_bestv1p0.html



Open Source Java API for LevelA IMS LD

While working on this project we had required a set of java api to manage the ims ld. Although we could get the level A xsd file from the ims website but we could not get the java api corresponding to the xsd files. The main requirement for these java api was that they should match the imslld xsd schema because it would simplify the understanding and usage of these api. There is one such api for ims qtilite specification at Universitat Pompeu Fabra, Spain [2] <http://gti.upf.edu/leteos/newnavs/jqtilite.html> but we could find none for ims ld.

So we decided to create these java api for imslld which was a fairly simple process. These java api have been auto generated using the ims ld version1.0 level A xsd file and the Open source castor [3] <http://www.castor.org/> . The good thing about these api is that since they exactly match the LevelA IMS LD xsd schema so one can map the various functions in api to the elements in xsd schema. This might help to understand and use these api to read and write ims ld. At present these api only manage the level A ims ld but we intend to extend them in future for level b and c ims ld. To help others to understand and use these api, Javadoc and a example java file demonstrating using these api to manage ims ld have also been created. These java api require jdk1.4.

These java api are open source and have been released under Apache 2.0 licens by Phosphorix Ltd. So they can be reused in other projects. Since these java api are reusable, so phosphorix would be delighted if other projects use them and would be happy to provide assistance if required.

Prototype for iosequence and IMS LD Bridge

As a starting point we have selected a simple learning design from the IMS website, created an equivalent learning design in ioportal and then compared the respective xml files. By doing so we intend to explore the possibility of interoperability between iosequence and ims ld. The whole process is divided into following three steps:

Step1: Select a ims ld xml file from ims website examples

We have taken the attached simple ld "imsld_boeing_imsmanifest.xml" from ims website to test the ioS-ims-Ld prototype. It's a level A ld, with few activities, activity structure and one play. To make things simpler we have modified this ims ld by deleting all but one activity.

Step2: Generate the iosequence ld xml file

Next we generated a similar learning design in iosequence. The screen shots for the generation of similar question in ioportal are present in the attached file 'ios 2 IMS LD Prototype Screen shots.ppt'.



Step3: Compare the ims and iosequence learning xml files

Next we compare the two learning designs and explore the possibility of interoperability between iosequence and ims.

The mappings are present in the file 'iosequence_IMSLD_Mapping.xls'.

S.No	IMS LD	IoS	Comment
1	Activity	Activity Step	For each activity step in iosequence we have a "BROWSE BUTTON FOR RESOURCE FILES".
2	Activity Structure	Sequence	
3	Roles	Roles	
4	Method-->role-parts*	Sequence-Role mapping	This element assigns the various activities to the roles. The activities and roles are picked up from the component section and linked together.

To demonstrate proof of concept, a prototype has been developed to create learning designs in iosequence. The prototype can generate ims learning designs.



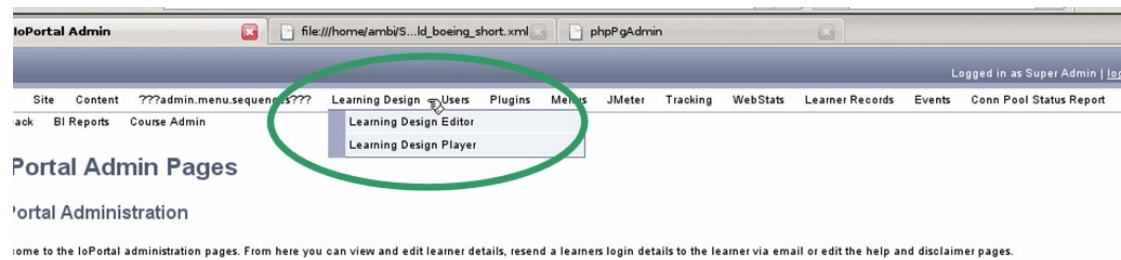
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Prototype Screen Shots



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Screen 1: Main Menu to Create Learning Design and Export as IMS LD, UOL.



Main Menu to Create Learning Design
and Export as IMS LD, UOL.

Learning Design Editor
Learning Design Player





Screen 2: Main Menu to Create Learning Design.

Learning Design Add Form

file:///home/ambi/S...ld_boeing_short.xml phpPgAdmin

Main Screen for LD Creation.

Logged in as Super Admin | Log out

Site Content ???admin.menu.sequences? Learning Design Users Programs Menus Semester Tracking WebStats Learner Records Events Conn Pool Status Report

ack BI Reports Course Admin

Learning Design Add Form

Title	ing Fuel Valve Removal simplified1	
Learning Objectives	learningobjectives1d1	
Prerequisites	prerequisites1d1	
Assign Roles to Activity Structures		
Activity Structure	Roles	Assign Roles
ochure		
Web		
ry_of_cv		
vey		
llet-NOS		
lts For Health - NOS		
nyms questions		
uestionnaire		
t survey		
n dev test		
tAmbiActivity1		
tAmbiActivity2		
tAmbiActivity3		
sequence_Choice_Question		
-introduction		
no1		
no2		
nold1		

Create Learning Design

System tray icons: Clock, Network, Volume, Learning application



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Screen 3: Main Menu to Create Learning Design.

The screenshot shows a web browser window with the title "Learning Design Add Form". The browser's address bar shows a local file path. The page has a navigation menu at the top with links like "Site", "Content", "Learning Design", "Users", "Plugins", "Menus", "JMeter", "Tracking", "WebStats", "Learner Records", "Events", and "Conn Pool Status Report". The main content area is titled "Learning Design Add Form" and contains several input fields and buttons. The "Title" field is filled with "ing Fuel Valve Removal simplified1". The "Learning Objectives" field is filled with "learningobjectives1d1". The "Prerequisites" field is filled with "prerequisites1d1". Below these fields are tabs for "Assign Roles to Activity Structures", "Activity Structure", "Roles", and "Assign Roles". The "Activity Structure" tab is selected, showing a dropdown menu with "mold1" and a text input field with "R-learner". There is a "Create Learning Design" button at the bottom. The bottom of the screen shows a taskbar with a clock and several application icons.

Learning Design Add Form	
Title	ing Fuel Valve Removal simplified1
Learning Objectives	learningobjectives1d1
Prerequisites	prerequisites1d1
Assign Roles to Activity Structures	
Activity Structure	Roles
mold1	R-learner
Assign Roles	
Create Learning Design	



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Screen 4: Main Menu to Export as IMS LD and UoL.

A screenshot of a web application titled "Learning Design". The interface includes a top navigation bar with links like "Site", "Content", and "admin.menu.sequences?". Below this is a table with columns "Learning Design Id", "Title", "View", "Edit", "Delete", "Export As IMS LD", and "Export As UoL". A row in the table shows a Learning Design ID and the title "Boeing Fuel Valve Removal simplified1". The "Export As IMS LD" and "Export As UoL" buttons in this row are circled in green. Two callout boxes, "Export as IMS LD" and "Export as UoL", have arrows pointing to these buttons. A green text box at the top of the screenshot reads "Main Screen to export LD as IMS LD and UoL." The bottom of the screenshot shows a Windows taskbar with a clock and several application icons.



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Screen 5: Exported IMS LD.

```
<learning-design identifier="d76dba5815d180580115d18992110001" level="a" uri="URI">
<title>Boeing Fuel Valve Removal simplified</title>
<learning-objectives>
<title>Learning objectives title</title>
- <item identifier="learningobjectivesld1">
<title>Learning objectives</title>
</item>
</learning-objectives>
<prerequisites>
<title>Prerequisites title</title>
- <item identifier="prerequisitesld1">
<title>Prerequisites</title>
</item>
</prerequisites>
<components>
- <roles>
<learner create-new="allowed" identifier="R-learner"/>
</roles>
- <activities>
- <learning-activity identifier="LA-fuel-valve-lesson-intro">
<title>LA-fuel-valve-lesson-intro</title>
- <activity-description>
<title>LA-fuel-valve-lesson-intro</title>
- <item identifier="I-fuel-valve-lesson-intro">
<title>Intro fuel valve</title>
</item>
</activity-description>
</learning-activity>
</activities>
</components>
<method>
- <play>
- <act>
- <role-part identifier="role-part-id">
<role-ref ref="R-learner"/>
<learning-activity-ref ref="LA-fuel-valve-lesson-intro"/>
</role-part>
</act>
</play>
</method>
</learning-design>
```

Exported IMS LD.



References

- [1] IMS, <http://www.imsglobal.org/learningdesign/>
- [2] Universitat Pompeu Fabra, Spain, <http://gti.upf.edu/leteos/newnavs/jqtilite.html>
- [2] <http://www.castor.org/>

Appendix

In this section we provide all other relevant materials.

Following are the attached files:

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8. **TestIMSLDAutoGenerateAPI.java**: This is an example file containing an example to show the usage of the Open Source Java API for managing levelA IMS LD.

About Phosphorix

In partnership with our clients, we have developed many web based 'people' centric systems. Phosphorix develop solutions for; personal development, professional / staff development, e-learning, tracking, content management, document management, messaging, data transformation, web site management, extranet, supplier management, intranet, corporate web, news, web services, mobile phone services and e-portfolio systems.



Phosphorix started out by providing software development services for corporate MIS [management information systems] which had to be available to intranet or the world wide web users via 'the web browser'.

Phosphorix continue to develop custom solutions primarily for use via any web browser. We specialise in implementing internet architecture, providing open standard web applications, from simple client / server applications to complex service oriented distributed network systems. Phosphorix provide solutions for the web sites, intranets, extranets and virtual private networks.

Phosphorix are advocates of secure, open source and open standard technologies.

Historically, our most notable customers have been Vodafone, Vivendi Universal, The University of Plymouth, John Moores University Liverpool, The University of Hertfordshire, Liverpool Hope University, Manchester Metropolitan University, The University of Bournemouth and the JISC.

We pride ourselves in adding a creative spark to the design and planning process and welcome new work at any stage of development.